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# Responsibility-Sensitive Safety: Practical Aspects for Industrial Adoption

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**Abstract**

Formal collision-avoidance models for SAE Level 4/5 automated vehicles have gained attention in recent years as an approach to provide safety assurance and reduce the burden of driving millions of kilometres to validate the automated driving system (ADS). Since Mobileye's introduction of Responsibility-Sensitive Safety (RSS) in 2017, it has emerged as a promising model because of its transparent formulas, open-source implementation, and widespread adoption. In this report, we provide a comprehensive overview of RSS, beginning with its s and considerations, along with the associated research. Additionally, we analyze the existing open-source implementation of RSS, including rule support, CARLA integration, and RSS adoption in the industry. We elucidate how RSS can be used in automated driving and review methods to balance between safety and traffic efficiency. Finally, we conclude with a summary of our findings and discuss several aspects of the framework important for its potential adoption within a company such as Stellantis.

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# 1 Introduction

Ensuring safety is a key prerequisite for the mass adoption of autonomous vehicles (AVs). Responsibility-Sensitive Safety (RSS) [1] was proposed by Mobileye/Intel as a mathematical safety model for AVs that formalizes implicit driving rules (for example, keeping a safe distance from a leading car in a car following driving s). Since its initial publication in 2017, the model has sparked considerable research interest, including efforts on the modifications of the original model, its calibration (determining model's parameters), etc. The framework has been implemented in Mobileye's products, including Mobileye, Chauffeur, and SuperVision, as well as in Baidu's Apollo stack [2]. Further, it has influenced standardization and regulation, via IEEE 2846 [3] and China ITS Alliance [4]. An open-source implementation of the framework exists [5]. While it stopped being maintained in 2025, the implementation had been previously integrated into CARLA, a widely used simulator for AVs.

In this report, we present RSS' basics and survey state-of-the-art related to RSS research and deployment. In particular, we focus on several important aspects of the framework including but not limited to, its adoption in the industry, its applications, calibration (model parameterization), open source implementations, and relationship to relevant standards. Finally, we draw on our findings to reflect on how RSS can be integrated into a company's processes, software, and toolchain (such as Stellantis'), and outline the benefits and limitations of the integration.

This report will be followed by a comprehensive comparison of three existing safety frameworks: RSS, Safety Force Field (SFF) [6], and Rulebooks [7].

The outline of the report is as follows. Section 2 presents the RSS safety model. Section 3 presents the existing industry adoption of the framework, as well as details of the existing open-source implementation. Section 4 discussed applications of RSS including but not limited to safety architectures and safety metrics. It also discusses RSS with respect to ISO 21448 [8] (SOTIF — Safety of the Intended Functionality) and IEEE 2846 [3] standards. Section 5 discusses efforts in balancing safety and traffic efficiency. Finally, the findings are summarized, and limitations and benefits discussed in Section 6. The report concludes in Section 7.

# 2 RSS: Description

## 2.1 Responsibility-Sensitive Safety

Responsibility-Sensitive Safety (RSS) is a formal model framework that translates common-sense driving obligations into mathematically verifiable rules for automated vehicles. In what follows, RSS refers to the formal model introduced

in "On a Formal Model of Safe and Scalable Self-Driving Car" by Shalev-Shwartz et al [1], which articulates safety obligations under explicit, reviewable assumptions and provides guarantees for collision avoidance through restrictions on vehicle kinematics. The framework is constructed to satisfy three aims:

1. It should be sound in the sense of matching societal norms and legal precedent.
2. It should yield a useful driving policy that does not confuse other road users with excessive caution.
3. It should be efficiently verifiable so that one can prove that an implementation conforms to the interpretation without long chain butterfly effects.

The authors call this aspirational result "utopia". If all agents follow the interpretation, there will be no accidents and, in practice, the autonomous agent never causes an accident and can compensate for reasonable mistakes by others. The framework is based on five common sense s:

1. Do not hit someone from behind.
2. Do not cut in recklessly.
3. Right of way is given not taken.
4. Be careful in limited visibility.
5. If an accident can be avoided without causing another one, then it must be avoided.

The authors also express that a mileage-based validation alone cannot demonstrate the extremely low failure rates expected for safety-critical systems. Guarantees must be based on a transparent mathematical argument that others can audit which complements fault-oriented analyses from ISO 26262 and is consistent with SOTIF's treatment of performance limits and reasonable assumptions.

## 2.2 RSS Core Constructs

RSS builds its formalism around a small set of concepts that consistently reappear in both longitudinal and lateral settings. The formulations of the framework are based on a *lane-centric coordinate system* that transforms the global coordinate system into one that consists of straight lanes of constant width. The $(d_{min})$ is the minimum separation between two agents that still guarantees no collision if each party acts within stated bounds. The $(t_0)$ is the first instant in time when the distance between two agents drops below the safe distance

and the framework switches from maintaining safety to restoring it. At the moment of danger and thereafter, the framework specifies a  which structures the actions that an agent can take to restore safety. The model yields closed-form safe-distance formulae and proper response envelopes where obligations are defined pairwise, one agent versus another. The framework does not prescribe a unique trajectory; instead, any trajectory that remains inside these envelopes and satisfies the distance requirements is acceptable.

The paper uses the pairwise property to argue that the rules scale to complex traffic scenarios without needing a single global resolution of all interactions. The same constructs also anchor responsibility assignment. If an accident occurs, responsibility attaches when an agent violates the proper response requirements that applied from the time of danger onward. The combination of safe distance, danger threshold, proper response, compositionality across pairs, and responsibility forms the backbone of the framework.

## 2.3 Parametrization

The RSS model leverages a set of parameters, for example, maximum acceleration or minimum deceleration of the ego vehicle. No numeric values for parameters are prescribed by the framework and the parameters are treated as placeholders that must be fixed for any given  (ODD) and then applied consistently in both the analysis of s and the corresponding proper responses. This parameterization makes the model explicit about timing and physical limits allowing proof that if the measured separation exceeds the computed safe distance at a given instant, then a collision can be avoided by agents that react and brake within the stated envelopes. Implementations can tie chosen values to empirical evidence and legal expectations without altering the formal structure. For illustration, parameters relevant for the longitudinal and lateral directions can be found in Table 1 and Table 2 respectively.

There is a substantial body of work on calibrating RSS parameters so that safety and traffic efficiency are balanced. A discussion of existing approaches is given in Section 5. Additionally, modifications of the RSS model have been proposed to improve traffic efficiency —discussion is in Section 5.

## 2.4 RSS Safety Rules

We introduce the safety rules for the safe longitudinal and lateral minimum distance. Both rules introduce a *safety condition* (i.e., a safe distance threshold, where violation of it signals the beginning of a dangerous situation) and a *proper response* that is sufficient to restore safety once the  becomes dangerous.

Beginning with the longitudinal direction, consider a vehicle following behind a leading vehicle in the same lane. Let $v_r$ and $v_f$ be their longitudinal veloc-

ities respectively. Given the parameters found in Table 1, the minimum safe longitudinal distance between the two vehicles is given by Lemma 2 from [1]:

$$d_{\min}^{lon} = \left[ v_r \, \rho + \frac{1}{2} \, \alpha_{\max}^{lon} \, \rho^2 + \frac{\left(v_r + \rho \, \alpha_{\max}^{lon}\right)^2}{2 \, \beta_{\min}^{lon}} - \frac{v_f^2}{2 \, \beta_{\max}^{lon}} \right]_+ \tag{1}$$

where $[x]_+ \coloneqq \max\{0, x\}$ (i.e., the ReLU function).

Table 1: RSS parameters used in the longitudinal direction

| Symbol | Unit | Agent | Description |
|---|---|---|---|
| $\rho$ | Time | All | The reaction time it takes for an agent to apply a control strategy after the start of a dangerous situation. |
| $\alpha_{\max}^{\text{lon}}$ | Acceleration | Ego | Maximum acceleration the agent can have during the reaction interval. Reflects the most aggressive admissible throttle input during reaction. |
| $\beta_{\min}^{\text{lon}}$ | Acceleration | Ego | Minimum braking deceleration that must be applied by the agent after the reaction interval to avoid collision. |
| $\beta_{\max}^{\text{lon}}$ | Acceleration | Other | Assumed maximum braking deceleration capability of the agent. |
| $\beta_{\text{min,correct}}^{\text{lon}}$ | Acceleration | Other | Minimum braking deceleration that must be applied by the agent after the reaction interval to avoid collision. Applies only for the opposite-direction case. |

The proper response for the longitudinal following scenario is defined as follows:

- During the reaction interval $[t_0, t_0 + \rho]$ the longitudinal acceleration of $c_1$ must not exceed $\alpha_{\max}^{lon}$
- After the reaction interval, $c_1$ must brake by at least $-\beta_{\min}^{\text{lon}}$ until a safe longitudinal situation is reached, after which it may apply any non-positive acceleration
- From the start of the dangerous situation, $c_2$ must brake by at least $-\beta_{\max}^{\text{lon}}$ until a safe longitudinal situation is reached, after which it may apply any non-negative acceleration

A similar minimum safe distance formula is derived for the longitudinal situation in which two vehicles $c_1$ and $c_2$ are driving in opposing directions towards each

other. Accounting for directionality in the frame of reference, consider that $c_1$ is driving in the positive direction with a velocity $v_1 \geq 0$ and $c_2$ is approaching in the opposite direction in the same lane with a velocity $v_2 < 0$. Given that the velocities of each vehicle after the reaction interval can be defined as $v_{1,\rho} = v_1 + \rho\,\alpha_{\max}^{lon}$ and $v_{2,\rho} = |v_2| + \rho\,\alpha_{\max}^{lon}$, the minimum longitudinal safety distance for vehicles travelling in the opposite direction is given by Lemma 3 from Shalev-Shwartz et al. [1], also given below,

$$d_{\min}^{\text{lon}} = \frac{v_1 + v_{1,\rho}}{2}\,\rho + \frac{v_{1,\rho}^2}{2\,\beta_{\text{min,correct}}^{\text{lon}}} + \frac{|v_2| + v_{2,\rho}}{2}\,\rho + \frac{v_{2,\rho}^2}{2\,\beta_{\min}^{\text{lon}}}. \tag{2}$$

The authors synthesize the longitudinal proper response for the opposite-direction situation as follows:

- During the reaction interval $[t_0, t_0 + \rho]$:
  - The longitudinal acceleration of $c_1$ must not exceed $\alpha_{\max}^{lon}$
  - The longitudinal acceleration of $c_2$ must not exceed $-\alpha_{\max}^{lon}$
- After the reaction interval:
  - $c_1$ must brake by at least $-\beta_{\text{min,correct}}^{\text{lon}}$ until it reaches a full stop, after which it may apply any non-positive acceleration
  - $c_1$ must brake by at least $\beta_{\min}^{\text{lon}}$ until it reaches a full stop, after which it may apply any non-negative acceleration

The lateral safety distance is formulated similarly to the longitudinal case with the addition of a lateral fluctuation margin parameter, $\mu$. In a normal driving situation, a vehicle will tend to have a small degree of lateral fluctuation as a result of negligible steering drift or jitter. The parameter $\mu$ intends to capture the amount of unintentional lateral fluctuation and incorporate it into the lateral minimum safety distance such that, at a minimum, both vehicles are always laterally separated by at least $\mu$.

To kinematically determine whether a lateral movement is in fact intentional or not, the authors introduce the notion of $\mu$-lateral velocity. Any lateral movement that stays within $\mu$ of a vehicle is considered unintentional and is prescribed a $\mu$-lateral velocity of zero. If, instead, a vehicle's lateral movement exceeds $\mu$, the movement is considered intentional and is prescribed a non-zero $\mu$-lateral velocity. The $\mu$-lateral velocity is only used within the proper response of a dangerous lateral situation, not in the safe distance calculation. Details about $\mu$ and the $\mu$-lateral velocity can be found in Appendix 7.

Table 2: RSS parameters used in the lateral direction

| Symbol | Unit | Agent | Description |
|---|---|---|---|
| $\rho$ | Time | All | The reaction time it takes for an agent to apply a control strategy after the start of a dangerous situation. |
| $\alpha_{\max}^{\text{lat}}$ | Acceleration | Both | Maximum lateral acceleration the agent can have during the reaction interval. Reflects the most aggressive admissible throttle steering input during reaction. |
| $\beta_{\min}^{\text{lat}}$ | Acceleration | Both | Minimum lateral braking deceleration that must be applied by the agent after the reaction interval to avoid collision. |
| $\mu$ | Distance | Both | Lateral fluctuation margin representing the allowable residual lateral movement before defining a dangerous situation. |

Given the lateral parameters found in Table 2, the minimum safe lateral distance between the two vehicles is given by Lemma 4 from Shalev-Shwartz et al. [1]:

$$d_{\min}^{lat} = \mu + \left[ \frac{v_1 + v_{1,\rho}}{2} \rho + \frac{v_{1,\rho}^2}{2\,\beta_{\min}^{\text{lat}}} - \left( \frac{v_2 + v_{2,\rho}}{2} \rho - \frac{v_{2,\rho}^2}{2\,\beta_{\min}^{\text{lat}}} \right) \right]_+ \tag{3}$$

where the lateral velocity of a vehicle after the reaction interval is given as $v_\rho = v \pm \rho\,\alpha_{\max}^{\text{lat}}$ dependent on the orientation of the vehicles in the frame of reference. The authors synthesize the proper response to a lateral dangerous situation as follows:

- During the reaction interval $[t_0, t_0 + \rho]$ both vehicles may perform any lateral action such that their lateral acceleration $\alpha_{\text{lat}}$ satisfies $|\alpha_{\text{lat}}| \leq \alpha_{\max}^{\text{lat}}$
- After the reaction interval:
  - Before either vehicle reaches a $\mu$-lateral velocity of 0, they must apply a lateral braking deceleration of at least $\beta_{\min}^{\text{lat}}$ in the direction away from the opposing vehicle
  - After either vehicle reaches a $\mu$-lateral velocity of 0, the may apply any $\mu$-lateral velocity in the direction away from the opposing vehicle

## 2.5 Evasive Maneuvers

The authors state that if all vehicles adhere to the proper responses found in the framework, then there will be no collisions. However, they also acknowledge

that this is not always the case by stating that the framework requires an agent to compensate for another agent's improper behavior with an evasive maneuver formulated as a proper response. Suppose two vehicles, $c_1$ and $c_2$, enter a dangerous situation at time $t_0$ as a result of $c_2$'s improper driving behavior. A prediction of $c_2$'s future kinematic state is required by $c_1$ to determine if $c_2$ will follow its proper response obligations. If it is detected that $c_2$ breaches its obligations, $c_1$ must execute an evasive maneuver which is defined as an executed trajectory that does not break its proper response obligations to any road participant and also ensures that no collision will occur between the two vehicles in question. The framework therefore allows a vehicle to do more than the minimum in a proper response (i.e. braking harder than required or steering laterally), but never less than required.

## 2.6 Road Geometry and Right-of-Way

In lanes, merges, roundabouts, and signalized or signed intersections, the framework assigns priorities to routes. Right of way is given, not taken and an agent on a non-priority route must maintain safe distances and apply a proper response to prevent a dangerous situation for agents on a priority route. Conversely, a priority agent may still be required to brake if distances and kinematics imply a preventable collision. The previously presented definitions of minimum safe longitudinal distance and the longitudinal proper response, as well as the minimum safe lateral distance and the lateral proper response, are generalized to handle two agents on two different routes that eventually intersect.

Signals and signs are handled within the route priority abstraction so that the same kinematic rules apply after priority is established. This prevents over conservatism in which non-priority traffic could force priority traffic to behave in an unnatural manner. The rules for signals and signs also reflect the RSS underlying principle of "right of way is given, not taken": if, for example, a vehicle has a green light at an intersection, it still has to brake to avoid a collision with a vehicle stuck at the intersection on the red-light route.

## 2.7 Unstructured Roads

The route geometry of roads with markings allows RSS to treat longitudinal and lateral situations separately for s. However, the lack of road geometry in s requires RSS to consider longitudinal and lateral situations at the same time through two dimensional trajectories. The framework prescribes the notion of reachable trajectory sets as a set of trajectories that an agent can take during a dangerous situation shaped as a polygon. Two primary trajectory sets are identified: the *braking* set and the *continue-forward* set. In this case, a situation is deemed dangerous once the braking sets between two users begins to overlap.

The braking set ($\tau_b$) describes the extent the agent can travel while braking.

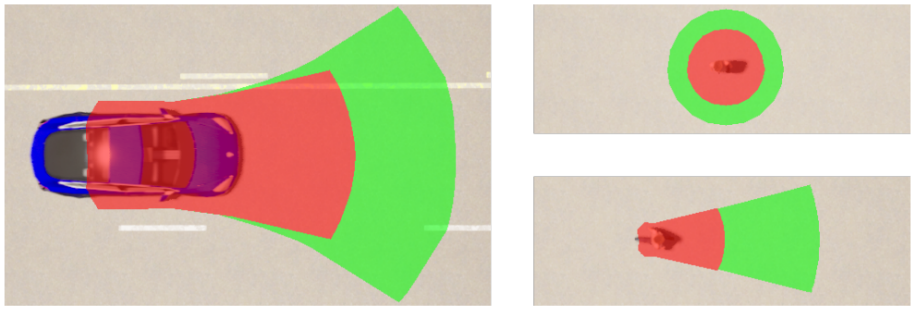

Figure 1: Trajectory sets for different road users describes as polygons for the braking set (red) and the continue-forward set (green). Left shows a vehicle, top right shows a standing pedestrian, and bottom right shows a moving pedestrian. [9].

The longitudinal extent of the braking set can be determined by a minimum and maximum travel profile. The maximum travel profile is described as accelerating by $\alpha_{\max}^{lon}$ during the response interval, and braking by $\beta_{\min}^{\text{lon}}$ thereafter. The minimum travel profile is described as braking by $\beta_{\max}^{\text{lon}}$ during the response interval, and braking by $\beta_{\max}^{\text{lon}}$ after the response interval as well. The lateral extent of the braking set is composed from the *change in yaw rate* parameter, which describes the rate and degree at which the agent will change its heading. In the case of a vehicle, during the response time, it is allowed to change its heading by the set maximum angular acceleration. After the response time it is assumed the the vehicle will continue on its curved trajectory in a circle. The continue-forward set ($\tau_f$) describes the extent the agent can travel if it were allowed to freely decelerate or accelerate. It is formulated similar to the braking set, however after the reaction interval, the agent can longitudinally accelerate at a rate of $\alpha_{\max}^{lon}$ for as long as it takes to brake to a complete stop at a rate of $\beta_{\min}^{\text{lon}}$. The trajectory sets can be decomposed into longitudinal and lateral components, however the evaluation of safe distances and proper responses requires the synthesis of both in unstructured settings.

Given two vehicles ($c_0$ and $c_1$), the situation in an unstructured setting between them is deemed safe so long as either condition given by Definition 22 in [1] is true. Condition 1 of Definition 22 states that if the braking set $c_1$ intersects the continue-forward set of $c_2$, so long as the braking set of $c_2$ does not intersect the continue-forward set of $c_1$, the situation is safe. The same applies in reversing the roles of $c_1$ and $c_2$. The mathematical definition is as follows:

$$\tau_{c_1,b} \cap \tau_{c_2,f} = 0 \text{ and } \tau_{c_2,b} \cap \tau_{c_1,f} \neq 0$$

$$\tau_{c_2,b} \cap \tau_{c_1,f} = 0 \text{ and } \tau_{c_1,b} \cap \tau_{c_2,f} \neq 0$$

Condition 2 of Definition 22 then states that so long as the braking sets of either car do not intersect, the condition is safe:

$$\tau_{c_1,b} \cap \tau_{c_2,b} = 0$$

If the situation is deemed unsafe through the violation of the conditions above, there are three responses that can be taken depending on the specifics of the condition violation: *continue forward*, *drive away*, or *brake*. To continue forward means to continue along a trajectory with no restrictions with respect to the continue-forward set, to drive away means to maneuver away from another agent given a specified heading range, and to brake means to slow down with respect to the braking set. The logic to decide which proper response to take is given in Figure 2.

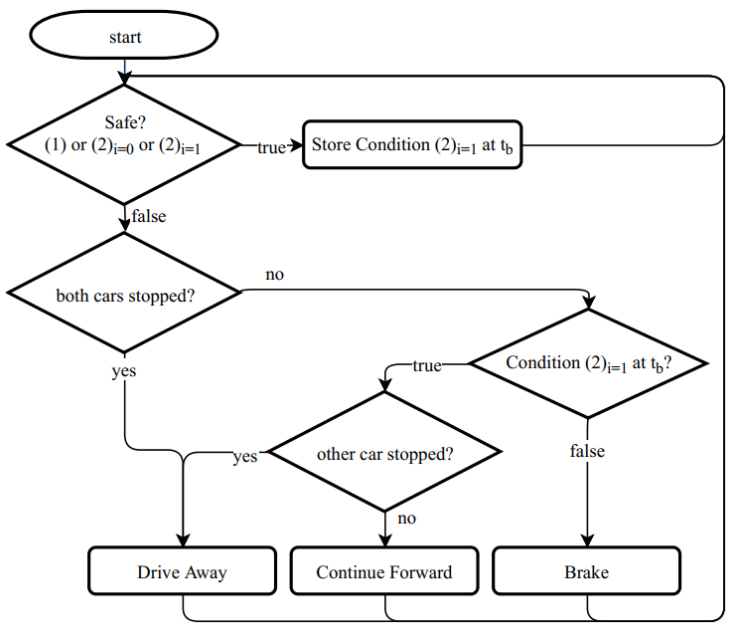


Figure 2: Decision logic for choosing the correct proper response regarding the conditions of Definition 22 [9].

## 2.8 Vulnerable Road Users

The parametrization and rules of RSS can be similarly extended to other s (VRUs), like pedestrians and cyclists. Each type of VRU needs its own set of unique parameters, rules on priority, and behaviour models to specify their

routes (structured and unstructured). For example, moving pedestrians in unstructured settings are given the same notion of trajectory sets, however unlike vehicles, they are modelled to continue in a straight line after the response interval of a dangerous situation. If the pedestrian is standing still, however, then the pedestrian can take any trajectory from the origin of its position (Figure 1). Though the framework mentions other road users, only the above example pedestrian behaviour is provided.

### 2.8.1 Parametrization of VRUs

Majority of the literature regarding the parametrization of RSS is related to vehicles. Though little work regarding determining parameter values for VRUs has been done, it is critically important to be able to accurately model the reasonably foreseeable behaviour of *all* types of road users for the framework. Pasch et al. [9] perform a literature review and an analysis of the inD dataset[1] to determine RSS parameters for vehicles, pedestrians, and cyclists. For VRUs specifically, they investigated literature consisting of a wide range of behaviour. For example, analyzing pedestrians at crossings and response times for attentive and distracted pedestrians. Figure 3 shows the summary of their findings regarding maximum accelerations and response times for each of their sources in the literature review. Note that, due to the wide range of use cases between literature, not every parameter can be extracted from a single source.

In their dataset analysis, they extracted the 99th percentile parameter values from the inD dataset for vehicles, pedestrians, and cyclists. The inD dataset contains drone-captured naturalistic trajectories of different road users traversing German intersections. Sine the dataset contains trajectories, they were able to extract the yaw rate and angular accelerations of the road users as well. Figure 4 contains the parameters they extracted.

They showed that the parameters for VRUs extracted from the dataset analysis are, on average, lower than the ones found in the literature review. This is due to the nature of the studies used in the literature review. Some parameters extracted described the *limits* of the road user behaviour, whereas the the dataset analysis contained *naturalistic* trajectories. Regarding reasonably foreseeable behaviour, it may be more prudent to analyze naturalistic behaviour rather than maximum extent behaviour. They also emphasize that parametrization, specifically lateral parameters, is highly dependent on the ODD.

## 2.9 Occlusions

Real traffic scenes frequently contain regions that are not directly observable because of geometry, parked vehicles, or imperfect perception. RSS addresses

[1] http://www.inD-dataset.com

| Ref | $a_{\max}$ [m/s$^2$] | $b_{\max}$ [m/s$^2$] | $b_{\min}$ [m/s$^2$] | $a_{\text{lat,max}}$ [m/s$^2$] | $\rho$ [s] |
|---|---|---|---|---|---|
| **Cars** | | | | | |
| [4] | 3.0 | 8.0 | 4.0 | - | 0.5 |
| [5] | 3.5 | 8.0 | 4.0 | - | 1.0 |
| [8] | - | 6.6 | - | 2.6 | - |
| [9] | 5.6 | 9.5 | - | 9.3 | - |
| [10] | 4.4 | 8.9 | - | 4 | - |
| [11] | - | - | - | 6.8 - 11.4 | - |
| [12] | 4.0 | 5.0 | - | 6.0 | - |
| **Bicycles** | | | | | |
| [13] | 1.5 | 3.0 - 4.8 | - | 3.6 | 1.0 - 2.5 |
| [14] | 0.8 – 1.2 | 2.6 | 1.5 | - | - |
| [15] | 0.71 | - | - | - | - |
| [16] | 0.8 | 0.8 | - | - | - |
| [17] | 0.8 | - | - | - | - |
| [18] | 1.4 | 3.3 | - | - | 1.3 - 2.3 |
| **Pedestrians** | | | | | |
| [19] | 2.7 | - | - | - | - |
| [20] | 0.23 | - | - | - | 0.33 - 2.87 |
| [21] | 0.78 | - | - | - | - |
| [22] | 0.906 | - | - | - | - |
| [23] | - | 4.66 | - | - | 0.68 - 1.0 |
| [24] | 1.44 | - | - | - | - |

Figure 3: Summary table showing the acceleration and response time parameters extracted from the literature review in [9].

| Type | $a_{\max}$ [m/s$^2$] | $b_{\max}$ [m/s$^2$] | $a_{\text{lat,max}}$ [m/s$^2$] | $h''_{\max}$ [rad/s$^2$] | $h'_{\max}$ [rad/s] |
|---|---|---|---|---|---|
| Car | 2.30 | 2.70 | 2.79 | 0.90 | - |
| Truck/Bus | 1.92 | 2.43 | 2.27 | 0.75 | - |
| Bicycle | 1.45 | 1.85 | 2.62 | 0.49 | - |
| Pedestrian | 0.76 | 0.66 | 0.79 | - | 0.50 |

Figure 4: Summary table showing the acceleration and angular rate parameters extracted from the d review inD dataset analysis in [9].

such cases by extending its distance-based semantics to interactions with hidden agents and by specifying how prediction and responsibility are evaluated when line of sight is absent. In this setting, RSS requires an agent to plan for reasonably expected behavior of other road users. The distinction between reasonable and unreasonable behavior is explicit in the treatment of both prediction and responsibility. Reasonable behavior assumes that other participants remain within the assigned parametric bounds and respect structural rules, such

as route priority. For unreasonable behavior, the paper identifies two situations that an agent is not required to anticipate.

The first is an unreasonable situation due to unreasonable speed. On any given route there are upper and lower limits on the expected velocity of a vehicle. For example, a vehicle on a residential road is not expected to travel at highway speeds. Accordingly, the framework prescribes defining upper and lower limits for longitudinal and lateral velocity based on scene structure and conditions. An agent must regulate its approach so that conflicts remain avoidable for other agents that could emerge within those limits, but it is not required to plan for arrivals whose speeds would exceed them. The second is an unreasonable situation due to improper behavior. RSS asserts that if all agents adhere to the restrictions of a proper response, collisions do not occur. A situation is therefore labelled unreasonable if it cannot be reached from some point in the past while all participants adhere to the RSS proper response. An agent is not required to plan for a situation if the only way to achieve the current state is by an agent violating the proper response at an earlier time.

These situations feed directly into the occlusion-aware proper response. An agent must satisfy the longitudinal and lateral safe distances with respect to all observed agents and must also plan for hypothetical agents that could emerge from occluded regions, but only under the reasonable behaviour model described above. When either unreasonable case would be necessary to make a collision possible, the obligation to plan for that collision does not apply. Conversely, an agent must avoid creating unreasonable situations through its own motion (e.g. exceeding prescribed limits on speed near occlusions) which would force others into an impossible history. Under these conditions the safety envelopes remain satisfied and the no-accident guarantee extends to limited visibility scenes when all participants comply with the occlusion rules.

### 2.9.1 Implementation of Occlusions

Gassmann et al. [10] provide an example implementation of occlusion handling in RSS using simulation. Their goal was to examine urban and highway scenarios to determine the trade-off between safety and traffic efficiency when assuming the reasonably foreseeable behaviour of occluded agents. Their setup consists of CARLA simulator alongside `ScenarioRunner` to run various `OpenSCENARIO` formatted scenarios. The RSS calculations are handled using a special node created for ROS which is bridged to CARLA using the `CARLA-ROS-Bridge`. A semantic LIDAR pipeline was used to identify occluded lane regions and visible objects. The architecture used for the study can be found in Figure 5.

To handle occluded vehicles, artificial agents were placed at the start of occluded lanes for each possible direction (Figure 6). Their reasonably foreseeable behaviour was described as always facing the nominal direction of travel for their lanes, and always driving within a prescribed speed interval (e.g. 0 to the

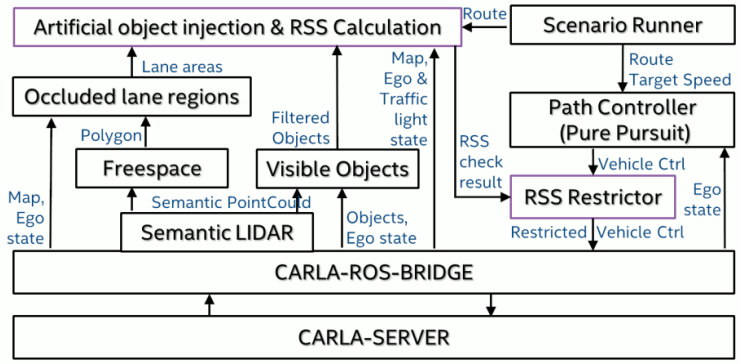


Figure 5: Experimental setup for the occlusion analysis [10].

speed limit of the road). These artificial agents are then treated as any other road agent, where the RSS safety envelope is calculated and the proper response is evaluated.

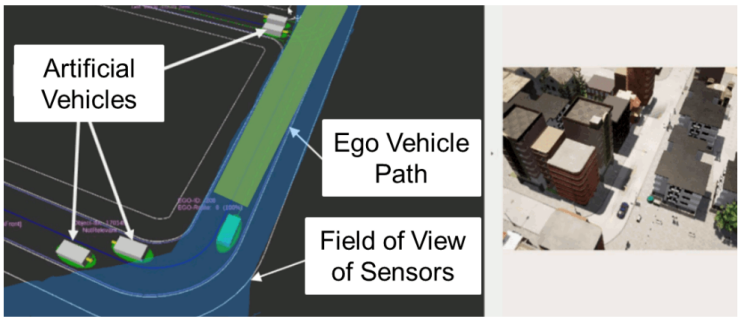


Figure 6: Positioning of artificial agents respective to the ego vehicle's (cyan) field of view (blue) [10].

The urban scenario examined in detail is depicted in Figure 7, where the ego vehicle (cyan) approaches an intersection and a building occludes another vehicle approaching the intersection laterally. The target metric for the experiment was the ego vehicle's travel time to a destination past the intersection.

The experiment was first run without any method of occlusion handling with varying right-of-way rules and RSS parameter sets. They found that without occlusion handling, there was always a collision between the two vehicles as the ego vehicle does not have enough time to apply its proper response once the other vehicle becomes perceivable. They then ran the same experiments, but inserted artificial agents with a speed interval between 0 $m/s$ and 8.33 $m/s$. The use of artificial agents prevented any collisions from occurring at all. The RSS

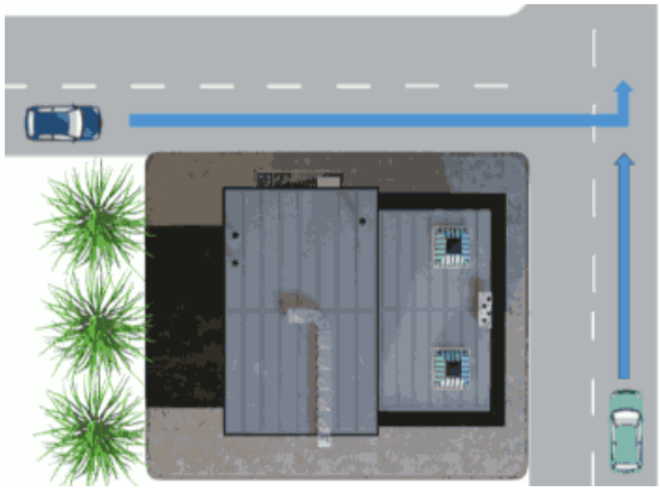

Figure 7: Urban scenario used for occlusion analysis. The ego vehicle (cyan) is unable to see occluded vehicle (blue) as they both approach the intersection [10].

parameter analysis also showed that a more aggressive parameter set (faster reaction time, stronger braking accelerations, etc.) led to longer ego destination travel times. With a more conservative parameter set, the ego vehicle maintains a safe distance such that it never has to brake hard enough to come to a complete stop. However, with a more aggressive parameter set, the ego vehicle would drive further into the intersection and closer to the other vehicle, resulting in the ego vehicle coming to a complete stop. Due to this, the agressive parameter set has a lower overall average speed compared to the conservative set.

The highway scenario examined in detail is depicted in Figure 9 where the ego vehicle (cyan), driving in front of two trucks back-to-back, attempts to quickly change one lane to its left where an occluded vehicle is approaching quickly from behind.

Similar to the urban scenario, the experiment was first run without any artificial agents where they found that once the occluded vehicle was perceived by the ego vehicle, it is too late to apply the lateral proper response and a collision occurs. Next, the same experiment was run by inserting an artificial agent with an assumed speed of 44.4 $m/s$ in the occluded regions of the trucks. They found that as the ego vehicle would initiate the lane-change, the artificial agent would cause the ego-vehicle to counter-steer back into its lane. Figure 10 shows the ego vehicle's lateral distance from the target lane's centre (blue) and if another agent is deemed dangerous. As the ego vehicle is changing lanes, at $t$ = 18.8 $s$ RSS flags the artificial agent (pink) as dangerous, causing the ego vehicle to counter-steer. As it is doing so, the visibility rearwards into the adjacent lane

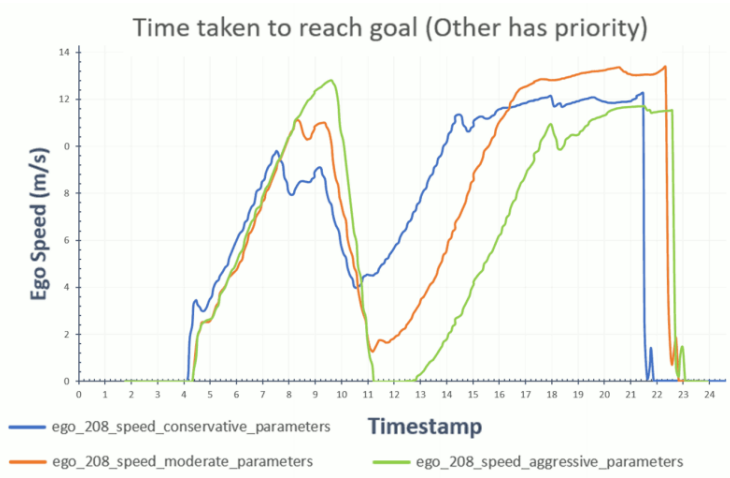


Figure 8: Speed versus timestamp plot results for the urban occlusion scenario showing the conservative parameters (blue), moderate parameters (orange), and aggressive parameters (green) [10].

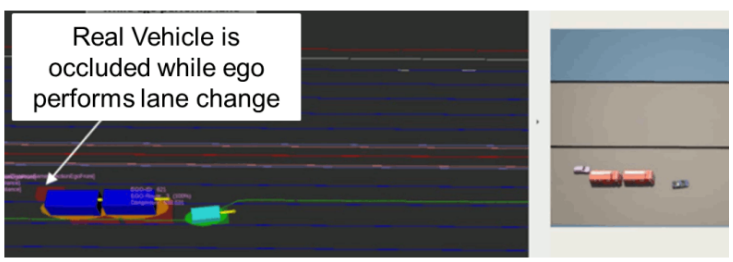


Figure 9: Highway scenario used for analysis. The other vehicle (red) is occluded by the trucks (blue) as the ego vehicle (cyan) attempts to change lanes. [10].

increases, allowing the ego vehicle to perceive the real vehicle (green) travelling in the adjacent lane at 50 $m/s$ at $t$ = 19.3 $s$, which RSS then deems dangerous. This prevents the ego vehicle from colliding into the real vehicle, allowing it to pass and then successfully make the lane change afterwards.

Another highway experiment was run where the ego vehicle is traversing an occluded corner at 36 $m/s$. They found that if they placed artificial objects in the lane ahead at a complete stop, conservative parameters would lead to much lower average speeds, and aggressive parameters would lead to overly aggressive braking. To overcome the difference in driving behaviour from the two parameter sets, the authors instead created two artificial agents at the same location. One artificial agent would be at a stand-still, which the ego vehicle would use a more aggressive parameter set to respond to. The other agent

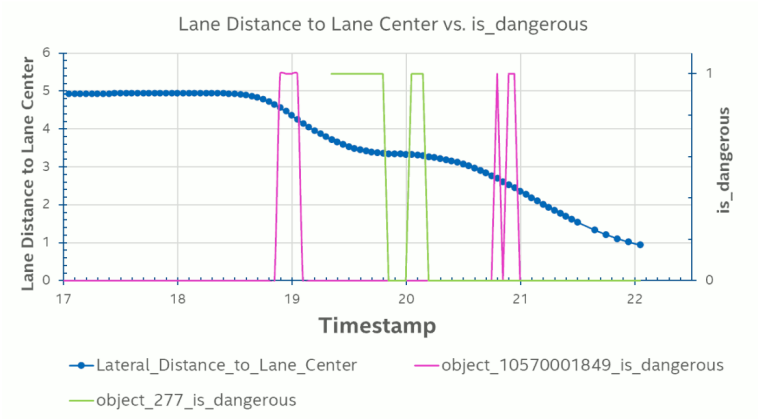


Figure 10: Lateral distance to the target lane centre (blue) and RSS danger identification for the artificial agent (pink) and the real vehicle (green) for the highway occlusion scenario [10].

would then have a speed of 13.88 $m/s$, which the ego vehicle would use a more moderate parameter set to respond to. This ensures that the ego vehicle is able to stop in time using the aggressive parameter set, while also ensuring that an acceptable overall average speed is achieved with the moderate parameter set.

Lastly, the authors analyzed the HighD[2] dataset to investigate reasonably assumable speeds that vehicles may travel at on the highway. They filtered the dataset to only include cases of collisions where leading vehicles were travelling 20 km/h faster than the rear vehicle, and only material damage to the vehicles was reported. A distribution of the speeds for these cases was plotted, showing that the total number of cases where "a leading vehicle (occluded or not occluded) is significantly slower than a rear vehicle is in total (overall speeds) less than 1.2%" [10]. Using a metric of $4\sigma$ and $5\sigma$ of the distribution, they stated that the reasonable foreseeable worst-case behaviour on a highway is vehicles travelling between 20 km/h to 70 km/h, not occluded standing vehicles.

# 3 Existing Support

This section describes RSS' open-source implementation `ad-rss-lib` based on our experience with the library as well as on the review of its GitHub repository and related literature (Section 3.1). Then, the framework adoption in the industry is surveyed in Section 3.2.

[2] `https://levelxdata.com/highd-dataset/`

## 3.1 Open-Source Implementation

The most prominent open-source reference is Intel's `ad-rss-lib`, a `C++` implementation of RSS [5]. Its primary objective is to implement the mathematical rules of RSS for use in autonomous driving software stacks as a reference for research. It allows for a continuous evaluation of an actor's relationship to other traffic participants, the detection of dangerous situations, and the computation of proper responses in the form of acceleration restrictions.

This section stems from a review of the library's GitHub repository[3], the library's documentation[4], as well as from experience using a version of CARLA simulator that was packaged with `ad-rss-lib`.

### 3.1.1 Library Overview

The library is structured to act in parallel to the planning module as a *doer-checker* in a sense-plan-act autonomous driving platform as seen in Figure 11. The overall goal of the library is to calculate the RSS safety envelope (minimum distances) from the extracted data and provide the proper response as acceleration restrictions. The planned action from the planning module is compared with the safety envelope to determine if the action is safe or not. If it is safe, then it is allowed. If the action is unsafe (violates safety envelope), then the proper response requirements must be followed. Note that it is up to the end-user to transform their perception data into the data structured required by the library, and to apply the proper response requirements on their own actuator commands. This is not the responsibility of `ad-rss-lib`.

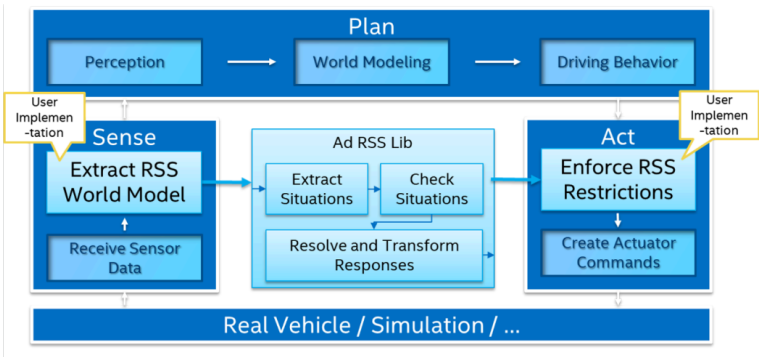


Figure 11: Integration of `ad-rss-lib` in an autonomous driving platform [5].

The library implements majority of the RSS rules and requirements except for a select few. First, *evasive maneuvers* in response to improper behaviour from

[3] https://github.com/intel/ad-rss-lib
[4] `https://intel.github.io/ad-rss-lib/`

other actors (Section 2.5) are not implemented. The library prevents the ego vehicle from creating dangerous situations with other actors by applying the appropriate acceleration restrictions. However, the library will not construct evasive restrictions on the ego vehicle if another actor maliciously tries to collide into it. Methods for handling *occlusions* are also not implemented as there is no mechanism to include hidden agents in occluded regions. Lastly, the library always assumes that any situation in an intersection contains a lateral conflict. If two vehicles are driving straight through an intersection in opposing lanes, though the situation is longitudinal, there will be a lateral conflict by default. The authors do not provide reasoning as to why this is the case. Table 3 summarizes the features of RSS that are and are not implemented in `ad-rss-lib`.

Table 3: Features and limitations of `ad-rss-lib`.

| **Implemented** | **Not Implemented** |
|---|---|
| Longitudinal conflicts (same and opposite direction) | Evasive maneuvers when needed due to other road users' improper behaviour |
| Lateral conflicts | Occlusions |
| Right of way | Intersections without lateral conflicts |
| Multi-lane road geometry | |
| Pedestrians | |
| Unstructured roads | |

The classification of situations into *constellations* is a central element of the library's operation. A constellation describes the spatial and kinematic pairwise relationship between two agents, classified into contexts such as same-direction travel, opposite-direction travel, or intersection approaches. In complex networks, a single agent may participate in multiple constellations with an ego agent, particularly where multiple connecting routes exist. The library processes each interaction between pairwise agents as an independent constellation, evaluates safety of each constellation, and merges the resulting constraints into a single output that can be applied by motion planning and control systems.

#### 3.1.2 Changes within `ad-rss-lib`

Several aspects of the library differ from the original RSS formulation to facilitate implementation in real-world environments and increase operability capabilities. The most notable change is the adoption of a *constellation-specific coordinate system*. Rather than operating in the lane-based coordinate system described in RSS, `ad-rss-lib` merges all lane segments relevant of a given constellation into a single road area coordinate space. This approach resolves po-

tential discontinuities in velocity and acceleration computations that can occur when road geometries change, such as on curves or variable-width lanes.

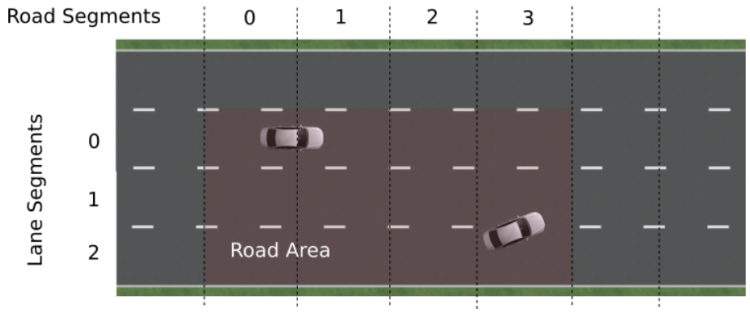


Figure 12: The *Constellation-Based Coordinate System*. The road area (shaded red region) uses only the road segments relevant to the situation [11].

The library also incorporates speed limit information into the acceleration assumptions used in the safe distance calculations to improve traffic flow efficiency. For example, in a dangerous longitudinal same-direction situation, RSS states that the ego vehicle accelerates at its maximum longitudinal acceleration during its reaction interval. If the vehicle was driving at the speed limit before the dangerous situation, assuming the vehicle will accelerate past that speed limit during the reaction interval may result in an overly conservative minimum distance formulation. The library changes the parametrization to assume that the vehicle will only accelerate up to a specific speed limit during the reaction interval instead.

Lastly, `ad-rss-lib` is capable of creating road boundaries to prevent the ego vehicle from maneuvering outside of its assigned road space, but it does so in a second-hand fashion. Rather than adopting a fundamental notion of "boundary", the library places virtual objects with their own set of parameters (no acceleration, minimum response time, etc.) along the edges of defined road areas through a naïve prediction of the ego vehicle's route. The boundary definition can be restricted in four modes: the ego vehicle's lane, the ego vehicle's lane plus the opposing lane, all drivable lanes, and the entire road area including shoulders.

### 3.1.3 Implementation Information

The documentation for the library contains a large amount of information about its design principles, specifically on what differs from formal RSS. It is a good starting point for understanding the library, however it lacks tutorials or examples that would help a user integrate it into their stack. It is recommended to refer to both the library documentation, as well as the source code documenta-

tion provided through `Doxygen`[5] when trying to use the library.

No specific requirements-based testing methods are mentioned within the documentation, though it provides a test coverage report that indicates over 98% of the lines of code, and over 97% of the functions are tested[6].

The transformation of sensor data to the required input `WorldModel` data struct is also not trivial. No documentation is available for exact instructions as to what is required, however the authors in [11] provide a high-level guide, and the library documentation[7] provides insights into what is required. In essence, the `WorldModel` struct contains two elements. First, a description of the ego vehicle's RSS dynamics is required. This includes all of the relevant RSS parameters for structured and unstructured settings determined by the user. Second, a list of constellations (e.g. longitudinal same-direction, opposite-direction, etc.) for each ego-object pair must be constructed by the user. This requires an understanding of the surrounding map (lane segments, traffic direction, etc.) and objects (positioning, velocity, etc.) to create a *Constellation-Based Coordinate System* for each ego-object pair (Figure 12). To aid in this effort, the authors of `ad-rss-lib` have also developed the `ad-rss-map-integration` library to perform the required data transformation for applications using `OpenDrive` map formats, such as CARLA driving simulator. However, for use with other driving map formats, the user must manually determine the RSS situations from their perception data and transform them to the appropriate format used by the library.

#### 3.1.4 CARLA Integration

`ad-rss-lib` was officially integrated into select past versions of CARLA for exploring the framework in a simulation environment. The most recent version of CARLA that includes `ad-rss-lib` as a packaged installation is version `0.9.14`, released December 2022. All versions of CARLA released thereafter do not contain `ad-rss-lib` as a packaged option, however the CARLA documentation states that `ad-rss-lib` can be added to a newer version of CARLA by linking and building from source. Though there is no confirmation on which specific versions of `ad-rss-lib` are compatible with the newer version of CARLA.

The implementation is split into two distinct modules within the CARLA framework. The `RssSensor` extracts the RSS world data from the ground truth simulation world data to perform the situation analysis and provide a proper response using `ad-rss-lib`. The `RssRestrictor` then applies the proper response on the CARLA vehicle controls if intervention is required.

The `RssSensor` is capable of determining the types of constellations between

[5] `https://intel.github.io/ad-rss-lib/doxygen/ad_rss/index.html`

[6] `https://intel.github.io/ad-rss-lib/coverage/ad_rss/`

[7] `https://intel.github.io/ad-rss-lib/ad_rss_map_integration/ConstructRSSConstellations/`

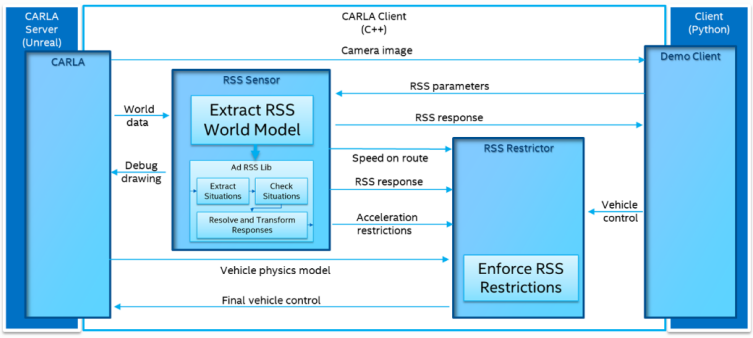


Figure 13: Interaction of the `RssSensor` and `RssRestrictor` in the `CARLA` architecture [12].

ego-object pairs through the use of the `ad-rss-map-integration` library. There is no need to manually determine the constellations as the simulator provides the ground truth world data and the `OpenDrive` map data for the `RssSensor`. It is to be noted that since the sensor module receives ground-truth data instead of data fed from a perception pipeline, it is not an accurate representation of how input data would be handled if `ad-rss-lib` was implemented in an autonomous driving stack.

Calculations within the `RssSensor` are also particularly slow when vehicles are driving in regions with complex road geometry (e.g. intersections or merges) since the module calculates a *Constellation-Based Coordinate System* for all the possible routes other vehicle can take. For example, if the ego vehicle and another vehicle are entering an intersection from opposing directions, it is not known if the other vehicle will drive straight, turn left, or turn right. Therefore the module will create a coordinate system for all three options and perform an analysis on each one. If there are many vehicles traversing the intersection, the computation slows down immensely as there are a great deal of contstellations to analyze.

The sensor is also extremely sensitive to errors or artifacts in `OpenDrive` maps. For example, some of the default CARLA maps have segments of road with an extremely small length to join different road networks or intersections. These small artifacts can lead to errors when calculating the coordinate system, resulting in incorrect safe distance calculations. Road boundaries would also fail frequently due to small map artifacts

There is also very little documentation regarding the implementation of the `RssSensor` and `RssRestrictor`. An example client running RSS on a manually controlled ego vehicle is provided with CARLA, however it does not contain much information to explain the logic of the code. A great deal of reverse engineering effort is required to apply the logic to other applications, such as applying the sensor module on an autonomous CARLA agent or other non-ego

agents.

### 3.1.5 Library Development

In August 2025, Intel announced that they would halt public maintenance of the `ad-rss-lib` repository. However, correspondence between the McMaster team and Dr. Bernd Gassmann of Intel in June 2025 about issues with the CARLA integration has provided insights about the on-going internal development of `ad-rss-lib`. First, support for Robot Operating Systems (ROS2) has been added to their internal library as a method to overcome compilation and linking difficulties with CARLA. Instead of using the `RssSensor` module in CARLA, they offload all of the RSS related calculations through a ROS2 bridge. This may be a reason why newer versions of CARLA do not have a packaged version of RSS in the releases. Second, they acknowledge two projects where their version of the RSS library was used: one to perform tests and evaluations for IEEE 2846, and the other to observe the Autobahn with many vehicles at the same time. These two projects were mentioned very briefly in the correspondence, with no additional detail. Both of these projects led to improvements to the library, such as patching flaws with road layout dependent calculations and adding multi-vehicle tracking support, which were not pushed to the open source implementation.

## 3.2 Industry Adoption

Mobileye has embedded RSS into the driving policy layer of its Mobileye Drive[8], Chauffeur[9], and SuperVision[10] platforms. A prominent case is the Volkswagen ID. Buzz AD, unveiled at IAA Mobility 2023, which explicitly utilizes an RSS based driving policy[11]. In March 2024, the Volkswagen group formalized a partnership with Mobileye to integrate the Chauffeur and Supervision platforms for level 2 and 3 autonomous driving, and level 4 for the Volkswagen Commercial brand[12]. In China, Baidu announced in 2018 that it would incorporate RSS into both its open source Apollo stack and its commercial Apollo Pilot program[13]. Similarly, Tier 1 supplier Valeo partnered with Mobileye to promote a global autonomous vehicle safety standard grounded in RSS principles[14].

[8] https://www.mobileye.com/solutions/drive/

[9] https://www.mobileye.com/solutions/chauffeur/

[10] https://www.mobileye.com/solutions/super-vision/

[11] https://www.mobileye.com/blog/volkswagen-id-buzz-mobileye-drive-iaa-mobility-2023/

[12] https://www.mobileye.com/news/automated-driving-volkswagen-group-intensifies-collaboration-with-mobileye/

[13] https://www.intc.com/news-events/press-releases/detail/145/baidu-to-integrate-mobileyes-responsibility-sensitive

[14] https://www.valeo.com/en/valeo-signs-an-agreement-with-mobileye-to-develop-a-new-autonomous-vehicle-safety-standard/

The influence of RSS extends into regulatory and standards setting activities. The China ITS Industry Alliance approved a technical requirement for automated driving decision making that draws directly from RSS[15]. At the international level, elements of RSS served as an early conceptual basis for the development of the IEEE 2846 2022 standard [13]. While IEEE 2846 does not prescribe RSS, its focus on the principle that safety policies should be defined in terms of explicit assumptions about road user behaviours aligns closely with the parametrization established in RSS.

# 4 Applications of RSS

Here, we outline the various ways RSS can be used for the benefit of automated driving. We take inspiration from Hasuo [14], who spoke in detail about four of the five applications presented in this section. In our review of the literature, we have found one additional application of RSS in the form of embedding RSS into AV algorithms, which we discuss in Section 4.3. Finally, in Section 4.6 we discuss the relationship between RSS, and SOTIF and IEEE 2846.

## 4.1 Formal Verification of ADS Safety

There are two major challenges in mathematically proving that every trajectory of the ADS complies with a list of RSS safety rules, and thus, provably assuring the safety of an ADS [14]. First, ADS controllers are very complex systems that include complex optimization algorithms and which often contain machine learning-based components. Thus, proving compliance of an ADS controller with RSS rules in the general sense is practically impossible. Moreover, to prove the safety of a given trajectory, one requires a comprehensive list of RSS safety rules which themselves are accepted as completely defining "safe behaviour". As far as we are aware, such a list does not exist because the number of driving scenarios that can be encountered in reality is very large.

Nevertheless, the behaviour can be verified against a given set of RSS rules in simulation, but the limitation imposed by machine learning-based components restricts verification to scenario-based testing, and therefore safe behaviour can not be claimed in general. But specifically in scenario-based validation activities, RSS provides a wealth of quantitative information on the behaviour of the ADS, which can be used to demonstrate compliance with standards such as SOTIF. This is discussed in more detail in Section 4.6.

[15] https://www.mobileye.com/technology/responsibility-sensitive-safety/

## 4.2 As a Safety Monitor

The structure of RSS Rules, being that of a condition and a response after condition violation, naturally lends itself RSS' application as a safety monitor, following a common design pattern such as the simplex safety architecture [14]. The simplex architecture adheres to an important principle in the design of safety-critical systems, which is the separation of safety and control systems [15]. Hasuo [14] presents the use of RSS in this prototypical safety architecture. In this application of RSS, the planning system contains an advanced controller (AC) and a baseline controller (BC) (Figure 14).

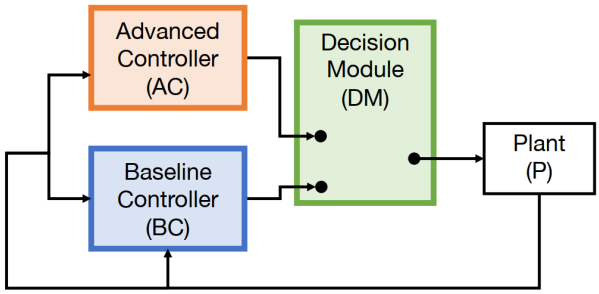


Figure 14: Diagram of the simplex architecture [14].

The advanced controller can be any high performance, complex controller that may pursue safety and/or performance (e.g., drive comfort, fuel efficiency, etc). The BC is a safety-certified controller that is concerned only with safety. The Decision Module (DM) contains logic that switches control authority between the two controllers. The DM will attempt to give authority to the AC as often as possible, but when it determines that the situation has become safety-critical, it switches control to the BC to execute a control action that will bring the vehicle into a safe state.

RSS can be used within a simplex architecture by using the RSS safety conditions as switching logic within the DM. When a minimum distance condition is violated, the DM will switch authority to the BC. The BC in this case will have the responsibility of implementing the proper response. Once the situation is safe (i.e., an RSS safety condition is no longer being violated), the DM switches authority back to the AC.

A major benefit of the architecture is that the safety of the ADS now depends on the simple, well-understood DM and BC and not on the complex advanced controller.

## 4.3 Embedded into an AV Algorithm

In contrast to approaches using RSS as a safety monitor, some studies have embedded RSS concepts into some part of the automated driving pipeline. Chai et al. [16] compute the RSS minimum distance once per control loop. If the current distance between the ego vehicle and the lead vehicle is less than $d_{\text{min}}$, decelerate by $a_{\text{min,brake}}$, otherwise accelerate with the acceleration requested by the ACC module. This use of RSS is very similar to the use as a safety monitor, but since the RSS safe distance and proper response are computed inside the control loop, there does not exist a separation of safety and control, and consequently, its benefits are lost.

Both N. Zhao et al. [17] and P. Lin et al. [18] integrate RSS safe distance calculations into a lane-changing path planner. N. Zhao et al. [17] introduces Lane Changing RSS (LC-RSS), which modifies a standard lane changing path planner by computing a 5th order bezier curve that satisfies RSS minimum distance constraints with respect to every vehicle in the environment. Naturally, this requires predicting the behaviour of nearby road agents, which in the paper, is done using machine learning algorithms. The study does not address what happens when behaviour prediction is incorrect. Moreover, P. Lin et al. [18] introduces a very similar lane changing path planner that also produces a path that does not violate the RSS minimum distances. This method also relies on predicting the behaviour of the vehicles in the nearby environment, but again, the authors do not address the consequences when behaviour prediction is inaccurate.

Due to the principle of separation of safety and control, **we believe RSS is best used as a safety monitor**—entirely separated from the automated driving planning and control software. Even as a safety monitor, safety assurance is conditional on reasonable parameterization. Embedding into an AV algorithm at best couples safety-critical software with the safety monitor, violating good safety-critical design principles. At worst, it relies on ML-based algorithms for behaviour prediction and makes the safety claims made by RSS hard to assure.

## 4.4 As a Safety Metric

Hasuo [14] remarks that the RSS safety rules can be used as a metric to evaluate the safety of a trajectory of a vehicle. For example, we can define an RSS safety metric which measures the degree to which a trajectory satisfies the RSS safety rules. The score can be determined by quantifying the difference between $d_{\text{min}}$ and the actual minimum distance, with a negative score given to trajectories that result in a violation of an RSS safety rule.

During our experimentation with the CARLA `ad-rss-lib` implementation, we measured the number of violations of the RSS safe longitudinal distance as a safety metric. This work was part of our scene graph work on measuring how

well this particular scenario (car following, with safe distance violated) is covered in a select dataset for the purpose of testing out the RSS proper response.

## 4.5 Attribution of Liability

The inventors of RSS clearly state their motivations for RSS in a whitepaper released by Intel/Mobileye in November 2017 [19], which was published just months after the seminal RSS paper [1]. This whitepaper meant for the general audience was wiped from the internet in 2019 for unknown reasons. In this whitepaper, Mobileye motivated RSS from two primary perspectives: provable safety assurance and economic scalability. The authors explain their rationale as to why provable safety assurance is important by reminding us that "complete avoidance of every accident scenario is unavoidable" while referencing Figure 15, stating that this scenario happens often and thus the goal of an AV should not be to avoid finding itself in this scenario.

Figure 15: The blue car can do nothing to ensure safety [19].

The authors state that if a collision occurs, "there will be an investigation, which could take months. Even if the human-driven vehicle was responsible, this may not be immediately clear. Public attention will be high, as an AV was involved". They then outline why RSS is useful, "Our solution is to set clear rules for fault in advance, based on a mathematical model. If rules are predetermined, then the investigation can be very short and based on facts, and responsibility can be determined conclusively". They present RSS as a mathematical model for determining fault in case of a collision.

Practically, this is done by placing liability on the road agent that did not comply with the proper response constraints.

The stated exception in the seminal RSS paper is that "when there are occlusions, there may be an accident with no clear responsibility", such as that given by Figure 16.

Taken verbatim from Sec. 3.10. of [1], "On one hand, the yellow car is allowed to assume that if there is a vehicle $c_2$ in front of the red car, then the red car is applying the proper response on it. So, the yellow car is not responsible for the

Figure 16: Occlusion scenario in which placing blame is non-trivial [1].

accident. It is true that the red car did not respond properly on the blue car, but the red car was not involved in the accident at all so it is not clear if we can blame it for an accident between the yellow and the blue car". Ultimately, the authors concede that placing blame is not as black and white as non-compliance with the proper response constraints.

Another complication arises when thinking about what "compliance with the proper response constraints" actually means. First, the proper response is only executed when a road agent has determined it is in an unsafe situation. This determination is based on the RSS minimum distance equations, which are based on parameters, including response time. If the parameter set is too liberal, a road agent may determine that it is in an unsafe situation at the point where a collision is inevitable. Placing liability then becomes murky, as it may result in rewarding a vehicle with the most conservative parameter set, which would never underestimate minimum distances, but practically would contribute to traffic inefficiencies.

## 4.6 On Standards Compliance

This section discusses some of the ways in which RSS can be used toward compliance with the SOTIF standard, and how RSS relates to the IEEE 2846 standard.

### 4.6.1 ISO 21448 – SOTIF

In general, the use of RSS with respect to the SOTIF standard falls into two categories, validation and operational activities. Figure 17 summarizes the SOTIF activity cycle, with the numbers indicating the relevant clauses of the standard.

In the summary that follows, we will refer to these numbered areas where we believe that RSS can be valuable.

For validation, RSS can help in providing supporting evidence for the satisfaction of the acceptance criteria for the reduction of residual risk due to unknown hazardous scenarios (area 11). RSS can be used in simulation activities to generate empirical evidence that the SOTIF activities have reduced the space of unknown hazardous scenarios toward an acceptable level. This can be evidenced,

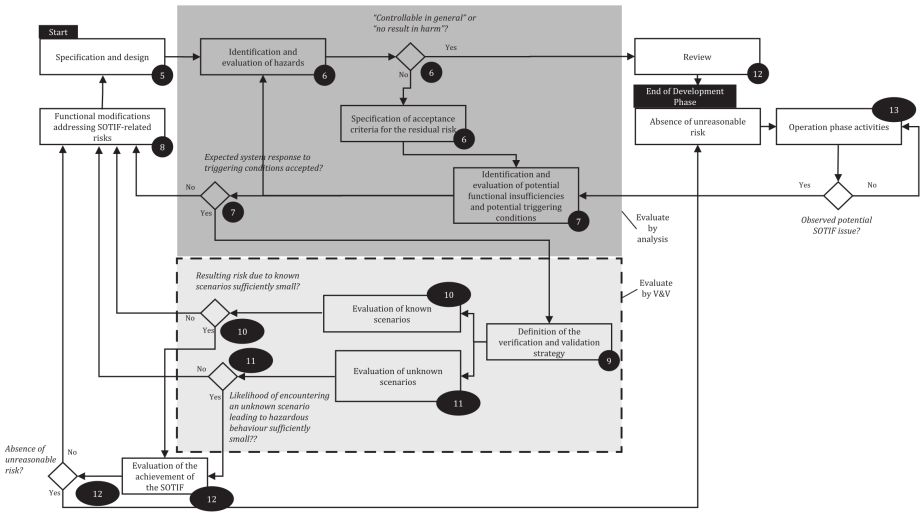


Figure 17: Overview of SOTIF activities.

for example, by a decrease in the number of RSS rules violations in simulation as the iterative activities of hazardous scenario identification proceed. A robust implementation of RSS will allow easy correlation between these violations and other elements of the implementation, specifically helping to point out functional insufficiencies to be addressed, as required by the standard. RSS can of course be used in area 10 as well in evaluating the acceptance criteria.

In the operational phase, RSS is helpful in two major ways. Mainly, the standard stipulates that, after SOTIF is achieved as a result of the recommended activities, constant monitoring during in-field operation be performed for the purpose of ensuring that the SOTIF is maintained. RSS is designed specifically to monitor the behaviour of the vehicle and to ensure that sets of rules are not violated by the control authority. Constant monitoring and enforcing of safety properties by RSS directly supports this requirement of the standard (area 13). Furthermore, RSS directly enables constant improvement to ML-based components when further unknown situations and scenarios are discovered during the operational phase. RSS rules violations can happen for diverse reasons, but two sources of particular interest to the standard are functional insufficiency caused by poorly-trained ML components, and reasonably foreseeable misuse. In order to draw benefit from these new discoveries (for instance, to re-train the ML-based components with the new information), and to differentiate them, all relevant information pertaining to the discovery must be retained (area 7). This information is available intrinsically in RSS and it can be correlated with the perception stream to differentiate the sources of violations, and help pinpoint ML-related deficiencies. In turn, this information helps create additional and accurate training data, feeding into the SOTIF activity cycle (area 8).

### 4.6.2 IEEE 2846

Earlier we mentioned that the relationship between RSS and the IEEE 2846 standard is very tight. In this section we expand on the standard in order to clarify this relationship. The two key elements of the standard, reasonably forseeable assumptions on the kinematic behaviour of other road users, and scenarios as a fundamental approach to achieving safe autonomous operation, are readily seen in RSS.

The IEEE 2846 standard addresses characterizing assumptions about the reasonably foreseeable behaviour of other road users that should be considered in safety models of AVs. The goal is to achieve a balance between safety and efficient traffic flow. The standard draws directly from natural human driving behaviour, where a driver's decisions with respect to other users of the road depend heavily on the driver's assumption of how those users will continue to behave (the *reasonably foreseeable behaviour*). But specifically in the standard, the assumptions are characterized as bounds on a set of variables describing the kinematic behaviour of other road users. The standard defines a minimum set of kinematic properties, including longitudinal and lateral velocity and acceleration, and heading angle and yaw rate. The reasonably foreseeable behaviour is encoded in the limits on these values, and these limits are attributed depending on driving situation. The standard does not specify these limits, it only states that they must be considered and assigned appropriate values, which depend on driving situation.

The standard suggests that these assumptions can be used in the definition of a "safety envelope" around the vehicle. RSS adopts this safety envelope approach. In the standard, the driving situations mentioned above are captured as minimal, or foundational, scenarios, with the intent being twofold. First, to be able to characterize driving situations in which the bounds on the kinematic behaviour of other road users can be easily understood as reasonably foreseeable; and second, to provide a minimal set of scenarios that can be composed to describe more complex driving situations. Whereas this composition is possible, the question of whether the assumptions on kinematic behaviour also compose in this simple fashion requires investigation. Similarly, the standard does not claim that the set of foundational scenarios is complete in terms of describing *any* driving situation, also requiring further investigation.

The standard also introduces attributes of safety-related models that can be verified/demonstrated by inspection as well as the ones that can be checked via validation testing. The former group contains 23 attributes including: support for reasonably foreseeable scenarios, focus on motion control, incorporation of assumption, support for emergency maneuvers, etc. The latter group contains five attributes including consideration of human violations of traffic laws and support for regional differences in behaviour. In fact, [20] considers how RSS (and other safety frameworks) embodies the attributes from the first group. To summarize the analysis, RSS directly incorporates assumptions.

# 5 Safety and Traffic Efficiency

Calibrating parameters that maintain safety assurance, while also not yielding such conservative behaviour that a vehicle equipped with RSS will become unusable, is a challenge. Some studies have investigated various methods of calibrating RSS parameters to balance safety and traffic efficiency, while others have taken the approach of modifying the RSS model itself. Studies investigating the former are discussed in Section 5.1; the latter is discussed in Section 5.2

## 5.1 Parameterization methods

### 5.1.1 Existing work

RSS, as well as the other safety frameworks are based on a parameterization of the reasonably foreseeable behaviour of other road users. The chosen parameterization(s) can be overly conservative, meaning that you assume other road users will drive more conservatively than they will in reality. This can result in the ego vehicle believing it is keeping sufficient distance to avoid collision when it is not. Conversely, assumptions on reasonably foreseeable behaviour can be overly aggressive, meaning that you assume other road users will drive more aggressively than they will. Though this may yield safe driving outcomes, traffic efficiency will suffer.

The literature on parameterizing safety frameworks heavily focuses on RSS, since the safe distance formulas are made very explicit in the seminal paper [1].

One approach to determine RSS parameters is to extract them from naturalistic driving data [21–23]. Xu et al. [21] extract safety-critical longitudinal following scenarios out of the Shanghai Naturalistic Driving Study. The extraction of this scenario was done using rule-based triggers (e.g., longitudinal deceleration exceeds -0.5 g). Then, each potential event was *manually* validated to confirm if it was indeed a longitudinal following scenario, and a near-crash or a crash. The authors calibrate relevant RSS parameters (e.g., $\rho, a_{\text{max,accel}}, a_{\text{min,brake}}, a_{\text{max,brake}}$) using a genetic algorithm to find the parameter set that maximized safety and minimized conservativeness. The safety metric used, Time-Integrated Time To Collision (TIT), represents the total time spent below a Time To Collision (TTC) of 4 seconds. The conservativeness metric was the difference between $d_{\text{min}}$ and the actual distance behind the lead vehicle. Though the concept of extracting RSS parameters from naturalistic driving data is legitimate, the study does not consider the fact that RSS parameters are not static — they should adjust to the type of road user that is being followed (a semi-truck will have a lower $a_{\text{max,brake}}$ than a sports car) and the ODD (e.g., road conditions, visibility, road classification). Additionally, manually extracting scenarios and identifying relevant ODD characteristics is time consuming.

We propose a solution to this in Section 5.1.2.

Wang et al. [22] also extract safety-critical scenarios from the Shanghai Naturalistic Driving Study using a similar methodology with the intent of studying pedestrian behaviour. A genetic algorithm was also used to find a parameter set for pedestrians to calibrate the RSS model for vehicle-pedestrian conflicts.

Nakamura et al. [23] take a similar approach, using naturalistic driving data, however they answer the question of what parameter ranges are common in the real-world such that an AV should be expected to handle them? They define a methodology to extract parameter ranges based on risk acceptance thresholds such as "probability of encountering a dangerous event is $10^{-6}$ times/year".

Some studies have used previously published studies that propose relevant values that can be used to compute reasonable parameter ranges [11, 24]. For example, Gassmann et al. [11] do a parameter analysis to compute $d_{\text{min}}$ for a sweep of parameters, using $\rho = 0.2$ s as a constant that was derived from a study that determined the response time of an AV. To determine the range of $a_{\text{min,brake}}$ and $a_{\text{max,brake}}$ to sweep, they refer to a study that gives the coefficient of friction for new and old tires on various road conditions, then compute using kinematic equations. This approach is fragile as it relies on idealistic kinematic equations and does not account for the kinematics of the lead vehicle.

Finally, Rodionova et al. [25] use an approach called Robustness Guided Falsification, which searches a parameter space to find a range of parameters that satisfy the RSS minimum distance equations. Parameters near the boundary of safe parameter space are considered more aggressive, while parameters further away from the boundary are conservative. Using this approach, the parameter space search would need to be recomputed for every ODD and for every lead vehicle to determine a parameter set that would lead to safety under all circumstances. We propose an automatic, scalable approach to extracting parameter ranges accounting for important factors that are relevant in determining parameter sets in Section 5.1.2.

### 5.1.2 Our Proposal for Determining Reasonably Foreseeable Behaviour: Scenario Extraction via Scene Graphs

A primary bottleneck in using naturalistic driving data to extract behaviour of road agents in the form of RSS parameters is automatically identifying the scenario [21, 23]. When this problem is solved, we can associate a set of parameters with a given scenario, then extract data points to obtain a distribution of parameters from naturalistic driving data. In this sub-section, we briefly propose an automatic approach to determining RSS parameter ranges, accounting for important factors such as the type of other road users and details about the ODD. Importantly, our approach is scalable and automatic, meaning that very little human input is needed and the method scales to many different scenarios.

We propose the use of Scene Graphs to automatically identify scenarios in a simulated or video dataset. A Scene Graph (SG) is a data structure that describes the semantics of a , such as entities and relationships between entities. First, we go through a dataset and generate an SG for every frame in the data. Then, we specify the semantics of a  using Linear Temporal Logic (LTL), and determine whether a sequence of Scene Graphs satisfies the scenario specification. This process allows us to determine every moment within a dataset that represents a scenario of interest. From this, we can associate a set of relevant parameters with a scenario (e.g., such as those identified in IEEE 2846), then store those parameters in a database. This method solves the more difficult, manual part of extracting parameters from naturalistic driving data — scenario extraction from an unlabelled dataset.

## 5.2 RSS Model Modifications

### 5.2.1 Existing Work

In addition to the studies that have looked into methods to parameterize RSS to balance safety and traffic efficiency, others have attempted to integrate efficiency into the RSS model itself.

First, Chai et al. [26] relax assumptions present in the RSS minimum distance equations to improve traffic efficiency. In the longitudinal safe following case, RSS makes assumptions about the behaviour of both the ego vehicle and the leading vehicle, such as the assumption that the ego vehicle will accelerate during the response time, $\rho$, parameterized by $a_{\text{max,accel}}$. In some cases, these assumptions can be relaxed as they are overly constraining. In a steady following state, it is unnecessary to accelerate, so the $\frac{1}{2}a_{\text{max,accel}}\rho^2$ term will increase the $d_{\text{min}}$ unnecessarily. Researchers at Tongji University collaborated with Intel Labs China to propose an improved RSS model [26] that divides the longitudinal RSS minimum distance calculation into three intervals.

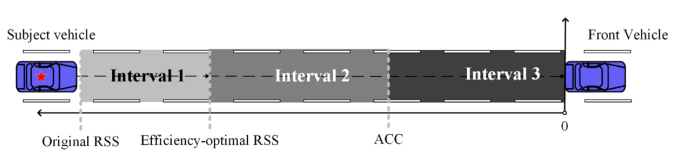


Figure 18: Proposed efficiency-optimal RSS model [26].

In their improved RSS model depicted in Fig. 18, the outermost safety distance interval is the original RSS $d_{\text{min}}$ calculation:

$$d_{\text{rss,margin1}} = \left[ v_r \rho + \frac{1}{2} \alpha_{\text{max, accel}} \rho^2 + \frac{\left(v_r + \rho \alpha_{\text{max, accel}}\right)^2}{2\beta_{\text{min, brake}}} - \frac{v_f^2}{2\beta_{\text{max, brake}}} \right]_+ \quad (4)$$

The next safety distance calculation, Efficiency-optimal RSS, relaxes the assumption that the ego vehicle will accelerate during the reaction time. In steady-state car-following, the new RSS safety distance is calculated by setting $a_{\text{max,accel}} = 0$:

$$d_{\text{rss,margin2}} = \left[ v_r \rho + \frac{v_r^2}{2a_{\text{min, brake}}} - \frac{v_f^2}{2a_{\text{max, brake}}} \right]_+ \quad (5)$$

Finally, the third efficiency-optimal RSS safety distance is defined by the lead vehicle suddenly decelerating at $a_{\text{max,brake}}$, then after a reaction time $\rho$, the ego vehicle decelerates at a *maximum* braking level $b_{\text{max,brake}}$, as opposed to the original RSS assumption that the ego vehicle brakes at a minimum braking level $a_{\text{min,brake}}$:

$$d_{\text{rss,margin3}} = \left[ v_r \rho + \frac{\left(v_r + \rho a_{\text{max, accel}}\right)^2}{2b_{\text{max, brake}}} - \frac{v_f^2}{2a_{\text{max, brake}}} \right]_+ \quad (6)$$

The authors introduce two versions of their improved RSS model: RSS1 and RSS2. The former triggers the RSS proper response when $d_{\text{rss,margin3}}$ is crossed and only imposes $a_{\text{max,accel}}$ when at the boundary of $d_{\text{rss,margin3}}$. RSS2 triggers the RSS proper response when $d_{\text{rss,margin2}}$ is crossed, and only imposes $a_{\text{max,accel}}$ when at the boundary of $d_{\text{rss,margin2}}$.

To evaluate the performance of the efficiency-optimal RSS model, the authors evaluated ACC with no RSS safety distances, RSS1, RSS2, and the unaltered RSS model, in simulation. They found that RSS1 did not see a sizeable improvement in safety compared with ACC, but it did improve traffic efficiency by approximately 10%. As expected, original RSS did not result in collisions, but also suffered in traffic efficiency. RSS2 was a middle ground, only resulting in collision when the lead vehicle deceleration was in the 99.9th percentile (i.e., $a_{\text{max,brake}} = -6$ m/s$^2$) and the time headway was 0.5 s or less. RSS2 also resulted in traffic efficiency levels that were in the middle of RSS1 and Original RSS. For a more detailed description of the results, we recommend reading the paper.

Other modifications to the RSS model include Li et al. [27] which uses a state-based system for the longitudinal following case. The authors define an approaching state, steady following state, and departing state, with each state having a corresponding $d_{\text{min}}$ formula. The steady following state formula,

$$d_{\text{min},1} = \left[ v_r \rho + \frac{v_r^2}{2a_{\text{brake}}} - \frac{v_f^2}{2a_{\text{max, brake}}} \right]_+ \tag{7}$$

shares an important similarity with Chai et al. [26]: both relax the assumption of ego vehicle accelerating during steady-state following. Additionally, Li et al. [27] replace $a_{\text{min,brake}}$ with $a_{\text{brake}}$, which is a braking level proportional to the current ego vehicle speed, defined as,

$$a_{\text{brake}} = a_{\text{max, brake}} + \frac{v_r}{v_{\text{max}}}(a_{\text{max, brake}} - a_{\text{min, brake}}) \tag{8}$$

where $v_{\text{max}}$ is the maximum speed of both vehicles. The evaluation of this efficiency-improved model was not done in simulation, but via a mathematical analysis. The analytical results showed that the improved model results in significantly lower safety distances than original RSS, but did not analyze how safety is affected.

Another study used an optimization approach to find RSS equations that were yielded lower $d_{\text{min}}$ values on average, however they did not prove the safety of their more "efficient" model, rendering this approach faulty as it negates the purpose of the provable safety guarantees that RSS makes.

### 5.2.2 An Error in the RSS Model

Koopman et al. [28] found that the longitudinal minimum distance equation (Equation 1) does not account for two vehicles from having overlapping positions *during* the braking response scenario — in other words, it does not prevent collision. Equation 1 is calculated assuming the minimum distance between vehicles happens at the final timestep in the scenario, however under certain circumstances, the minimum distance between vehicles can happen during the braking response.

Consider a rear vehicle with superior braking capabilities approaching a lead vehicle with inferior brakes at a high relative velocity. The rear vehicle may collide with the lead vehicle after the response time (i.e., while braking), but Equation 1 only guarantees that the vehicles will not be in the same position after the braking scenario has ended. The authors propose a modified version of the longitudinal safe distance equation, which is only applicable when $v_r > v_f$ at the end of the response time and when the rear vehicle has better braking capabilities than the lead vehicle (i.e., $a_{\text{min,brake}} >> a_{\text{max,brake}}$).

# 6 Discussion

We summarize the most important findings of our investigation of RSS, including RSS' major benefits and challenges regarding its potential adoption within an automotive company such as Stellantis.

- RSS is a popular safety framework, employed by Mobileye in their devices, and used by Baidu in their Apollo software stack. The framework has also influenced the development of standards and regulations, and is supported by several other organizations [4].

- Compared to other existing safety frameworks, such as Safety Force Field (SFF) [6] or Rulebooks [7], there is a significant body of research interest and results behind RSS, largely by Mobileye/Intel and academia. The research focuses on several topics: maneuvers not covered by the vanilla RSS (e.g., swerving), modifications of the vanilla RSS model to balance safety and traffic efficiency, as well as parameterization of the model for the same purpose, etc.

- RSS has many applications, including in safety architectures/monitoring, for attribution of liability, for formal verification, as a safety metric, and as part of path planning algorithms (e.g., RSS minimum distances being integrated as hard constraints in a lane changing planner); see Section 4 for details. Among the RSS applications, the most attractive – and increasingly common [14] – is its deployment in a safety architecture, where, for example, RSS' application in a traditional simplex architecture (see Section 4.2) allows for a clean separation between safety and control, enabling lower development, verification, maintenance, and certification efforts. For example, the widely touted benefit of RSS, formal verification, can now be applied on the relatively simple monitor and safety (baseline) controller (see Section 4) only. This is in stark contrast to the case where RSS is integrated in path planning algorithms, where formal verification would be very challenging given the complexity of the algorithms. We have also found RSS' application as a safety metric useful. In particular, we used it to count instances of the ego vehicle nearing too close (RSS minimum distance being violated) to a leading vehicle in car following scenarios identified in a CARLA dataset. While we leveraged the metric in simulation, where it can be useful as a safety performance indicator (SPI) early in the development cycle, the metric can be implemented on-board or off-board and evaluated online during driving.

- For a company looking to integrate RSS into its development process, reusing the existing `ad-rss-lib` library is an option. Our investigation of the library concluded several benefits and limitations of its use.

  Benefits are:

  - Reusing an open-source solution that was put forward and supported by Intel presents a cost-effective alternative to developing from scratch as the library already implements majority of RSS scenarios.

  Limitations are:

  - The library is missing some important use cases for AVs, including, most notably, occlusions as well as evasive maneuvers in response to improper behaviour from other road users. The latter is a responsibility of RSS given RSS' fifth principle: "If an accident can be avoided without causing another one, then it must be avoided."
  - There is no indication that the library has been developed in accordance with either ISO 26262 or another (national or international) standard. According to ISO 26262, this would mean that the reuse of the library would require efforts to argue the library's fitness for use in a safety critical application such as automated driving. One potential avenue would be making a proven in use argument; this is an unlikely option in the `ad-rss-lib` case as the required data would not be available as well as the fact that the library has been continuously changing from 2018 to 2025. The other option would be to qualify the library according to ISO 26262, SOTIF, and ISO/PAS 8926:2024 (Use of Pre-Existing Software Architectural Elements)—no small task by any means.

- The parameterization of RSS is challenging and should reflect a fine balance between safety and traffic efficiency. Several publications propose different parameter values for different environmental conditions as well as different methods to calibrate the model. While IEEE 2846 is a welcome effort on standardizing assumptions for safety models for AVs, it advises only on the kinematic properties that should be considered in AV safety models, failing to propose concrete numerical values for ranges of these properties. Unfortunately, in general, a consensus is still lacking in the industry when it comes to numerical values of boundaries of the model's parameters.

- Regarding occlusions, there is little to no discussion in the literature on methods of parameterizing the behaviour of phantom vehicles. It is possible to adopt the typical human approach to some forms of occlusion by analyzing video for different perspectives of an occluded scene. For instance, it is impossible to know what lies around a heavily occluded tight turn in the road V2X technology can help to some extent with this, but there is another option that relies on historical knowledge. A human driver will approach such a situation with caution if it is known that that particular turn is usually dangerous (e.g., due to children often playing in the road at that spot). If driving video of the same scene from a perspective that offers an unoccluded view is analyzed, a probabilistic understanding of the level of danger of that particular occluded scene can be developed,

and an autonomous agent can take that probability into account in approaching the scene with caution, which manifests as the behaviour of possibly occluded road agents. This mirrors the approach that humans take—relying on prior knowledge. These probabilities can then be further utilized in safety analyses of the autonomous features in determining acceptance criteria.

- RSS was introduced as a way of addressing the excessive validation burden on autonomous vehicles [1], however, quantitatively, it is unclear how much the use of a formal safety model reduces validation hours. This is an interesting open question that will hopefully be tackled by the research and standardization communities.

# 7 Conclusion

This report presents RSS based on the review of the existing literature, as well as on our experience with a framework's implementation in CARLA, a widely used AV simulation tool. We present the basics of the framework as well as its most important aspects such as potential applications, trade-offs between safety and traffic efficiency, and relationship to SOTIF and IEEE 2846. Finally, we summarize our thoughts to present some important factors to be considered about the potential integration of the framework into a company's automated driving.

This report will be followed by a report on the comparison of three major safety frameworks, RSS, SFF, and Rulebooks. The first draft of the comparison report is expected to be released to Stellantis before December 5th, 2025.

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# Appendix A: Lateral Fluctuation Margin and Velocity

As stated in Section 2.4, the lateral fluctuation margin, $\mu$, is meant to capture any unintentional lateral movement, while the concept of $\mu$-lateral velocity describes whether or not a lateral movement was intentional or not. Consider Car 1 in Figure 19 with a point fixed to its front-right corner an arbitrary lateral location $l$. The arrow describes the point's movement through time and the lateral fluctuation margin extends $\frac{\mu}{2}$ out from either side of the vehicle.

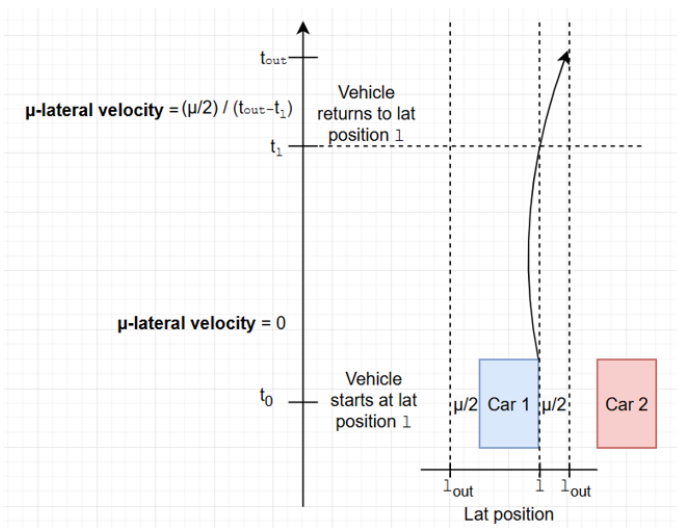


Figure 19: A depiction of $\mu$-lateral velocity.

As Car 1 travels forward from a starting time, $t_0$ to a time in the future, $t_1$, the point's lateral position stays within the bounds of $\left[-\frac{\mu}{2}, -\frac{\mu}{2}\right]$. In this case, the vehicle is prescribed a $\mu$-lateral velocity of zero since the lateral movement stayed within the fluctuation margin, and the movement is deemed as unintentional.

From time $t_1$ to another time in the future, $t_{out}$, the vehicle's movement causes the lateral position of the point to move to $l_{out}$, a lateral point $\frac{\mu}{2}$ away from the original position. In this case, the vehicle is prescribed a non-zero $\mu$-lateral velocity since the lateral movement exceeded the fluctuation margin, and the movement is deemed intentional. The $\mu$-lateral velocity is formulated as follows:

$$\frac{l_{out} - l}{t_{out} - t_1}$$